

Lecture 2 - Boolean Algebra

CIS-220

What did we go over last week?

- Course Policies
- Exam schedule
- Quiz schedule
- Assignments
- Introduction to Logic
- Truth Tables
- Course Overview

Logical Conjunctions

- Or
- And
- Not

All three of these logical Operators can be derived from every computer program

How do we turn our linguistic objectives into logistic answers and conditions?

p = years of service

q = bacholors

I want a candidate to have 15 years of experience and a bachelor's

→ I want a candidate to have 15 years of experience and a bachelor's

p = 15 years of experience

q = has a bachelor's

Does the candidate have 15 years of experience?

- Yes
- No

Does the candidate have a bachelor's?

- Yes
- No

if p and q are true, then it's true that I want to hire them.

$$p \wedge q = 1$$

This is boolean algebra

What is Boolean Algebra?

- Is the algebra of logic
- Developed by George Boole in 1854
- Works with two logical values
 - True
 - False
- Uses logical operators (logical conjunctions)
 - AND
 - OR
 - NOT
 - Conditional
 - Biconditional
 - Exclusive OR
- Boolean algebra lets us change the form of a logical expression without changing its meaning.

Boolean Operators

- AND: $p \wedge q$
- OR: $p \vee q$
- Not: $\neg p$

If Statements in Boolean Algebra

How do we represent them?

$p \rightarrow q$

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Example

Let:

p = "It is Tuesday"

q = "I will be at the party"

Then:

$p \rightarrow q$

Means:

"If it is Tuesday, then I will be at the party."

The only way this promise is false is if it is Tuesday and I do not show up.

“Let-then-means” frame work:

Let:

$q = x$

$p = y$

Then:

$q \wedge p = x \text{ AND } y$

$q \vee p = x \text{ OR } y$

$\neg q = \text{not } x$

Means:

It is x and y

Its is x or y

This is not x

Example

Let:

$p = \text{"It is raining"}$

$q = \text{"It is cold"}$

Then:

$p \wedge q = \text{"It is raining AND it is cold"}$

$p \vee q = \text{"It is raining OR it is cold"}$

$\neg p = \text{"It is NOT raining"}$

Logical Equivalence

$$P \equiv Q$$

- P and Q always have the same truth value
- They can look different
- They still represent the same logical result

Example

$$\neg(\neg p) \equiv p$$

- If we say:
 - It is not true that it is not raining
- This has the same logical meaning as:
 - It is raining

Double Negation Law

$$\neg(\neg p) \equiv p$$

- Applying NOT twice returns the original value
- Similar to saying "not not"

p	$\neg p$	$\neg(\neg p)$
T	F	T
F	T	F

Law of Excluded Middle

$$p \vee \neg p \equiv T$$

A proposition must be either true or false.

Example:

The computer is on OR the computer is not on.

One of these must be true.

Law of Contradiction

$$p \wedge \neg p \equiv F$$

A proposition and its negation cannot both be true.

Example:

The computer is on AND the computer is not on.

This cannot be true.

Identity Law

Identity Laws

$$T \wedge p \equiv p$$

$$F \vee p \equiv p$$

T = True

F = False

Adding a value that cannot change the result leaves p unchanged.

Example: $T \wedge p \equiv p$

Let:

p = "The computer is connected to the Internet."

Then:

$T \wedge p$

Means:

"True AND the computer is connected to the Internet."

Since T is already true, the result depends entirely on p .

Idempotent Law

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

Repeating the same proposition does not change its value.

Example:

It is raining AND it is raining.

Commutative Laws

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

The order of propositions does not change the result.

Example

It is raining AND it is cold.

It is cold AND it is raining.

It is raining OR it is cold.

It is cold OR it is raining.

Associative Laws

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

Changing the grouping does not change the result

Useful when working with several AND or OR operations

Instead of worrying about:

$$(p \wedge q) \wedge r$$

or

$$p \wedge (q \wedge r)$$

we can often simply think of:

$$p \wedge q \wedge r$$

Distributive Laws

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

and

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Let:

$a = \text{"This card is an ace"}$

$s = \text{"This card is a spade"}$

$c = \text{"This card is a club"}$

Then:

$a \wedge (s \vee c)$

Means:

This card is an ace and is either a spade or a club.

This is equivalent to:

$(a \wedge s) \vee (a \wedge c)$

This card is the ace of spades OR the ace of clubs.

DeMorgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Easy Rule to Remember:

When NOT moves inside

→ **AND becomes OR**

→ **OR becomes AND**

Each proposition is negated

Statement:

A raven is large and black.

Means:

The bird is large AND black.

Let:

$p = \text{"The bird is large"}$

$q = \text{"The bird is black"}$

Then:

$p \wedge q$

What does $\neg(p \wedge q)$ Mean?

The bird could be:

- Not large
- Not black
- Not large and not black

Therefore:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Negating an "If" Statement

Recall:

$$p \rightarrow q \equiv \neg p \vee q$$

Therefore:

$$\neg(p \rightarrow q)$$

becomes:

$$\neg(\neg p \vee q)$$

Using DeMorgan's Law:

$$\neg(\neg p) \wedge \neg q$$

Using Double Negation:

$$p \wedge \neg q$$

Final Result

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

Simplifying

Boolean algebra allows us to simplify expressions step by step.

Start:

$$p \wedge (p \rightarrow q)$$

Replace the implication:

$$p \wedge (\neg p \vee q) \text{ (These are logically equivalent: } p \rightarrow q \equiv \neg p \vee q)$$

Distribute:

$$(p \wedge \neg p) \vee (p \wedge q)$$

Contradiction:

$$F \vee (p \wedge q)$$

Identity:

$$p \wedge q$$

Result

$$p \wedge (p \rightarrow q) \equiv p \wedge q$$